

ENVS 193DS Homework 2

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2026-05-07

Part 1: Tasks

```
# loading in packages
library("tidyverse")
library("janitor")
library("effsize")

# creating new object called urchins, storing data
urchins <- read_csv("urchin-experiment-data.csv")

# creating new object called rides, storing my data
rides <- read_csv("rides.csv")
```

Part 2: Problems

Problem 1. Raptors at the bird refuge

- I collected discrete numerical data because there can only be an integer value of birds in the refuge.
- Standard deviation is a better description of the spread in Red-tailed Hawk count because it represents the variability of each data point from the mean.

```
# saving the raptor counts as an object called raptors
raptors <- c(1,3,2,0,1,1,3,2,0,0,1,1)
# calculating sd of raptors
sd(raptors)
```

```
[1] 1.05529
```

The standard deviation of raptor counts is 1.1 hawks.

- c. Standard error is a better description of the precision in Red-tailed Hawk count since it represents how accurate the sample mean is compared to the population mean.

```
# calculating se of raptors
sd(raptors) / sqrt(length(raptors))
```

```
[1] 0.3046359
```

The standard error of raptor counts is 0.3 hawks.

Problem 2. Urchin gonad weights

a.

```
# create new object from urchins
urchins_clean <- urchins |>
# clean column names
  clean_names() |>
drop_na() |>
# change c and n to be full words
mutate(experimental_treatment = case_when(
  # fill in Control to replace c
  experimental_treatment == "c" ~ "Control",
  # fill in Nutrient-enriched to replace n
  experimental_treatment == "n" ~ "Nutrient-enriched"
)) |>
# select experimental_treatment and gonad weight columns
select(experimental_treatment, weight_g)

# display 10 random rows
slice_sample(urchins_clean, n = 10)
```

```
# A tibble: 10 x 2
  experimental_treatment weight_g
  <chr>                <dbl>
1 Control                42
```

2	Nutrient-enriched	38.9
3	Control	33.4
4	Control	24.4
5	Control	25.5
6	Control	41.1
7	Nutrient-enriched	31.6
8	Control	39.6
9	Nutrient-enriched	27.3
10	Control	29.3

b.

```
# create new object from urchins_clean
urchins_summary <- urchins_clean |>
  group_by(experimental_treatment) |>
  summarize(
    mean = mean(weight_g),
    n = length(weight_g),
    sd = sd(weight_g)) |>
  ungroup() # ungroup the data frame

urchins_summary
```

```
# A tibble: 2 x 4
  experimental_treatment mean     n    sd
  <chr>                <dbl> <int> <dbl>
1 Control              26.8   42  8.67
2 Nutrient-enriched    31.7   35  3.91
```

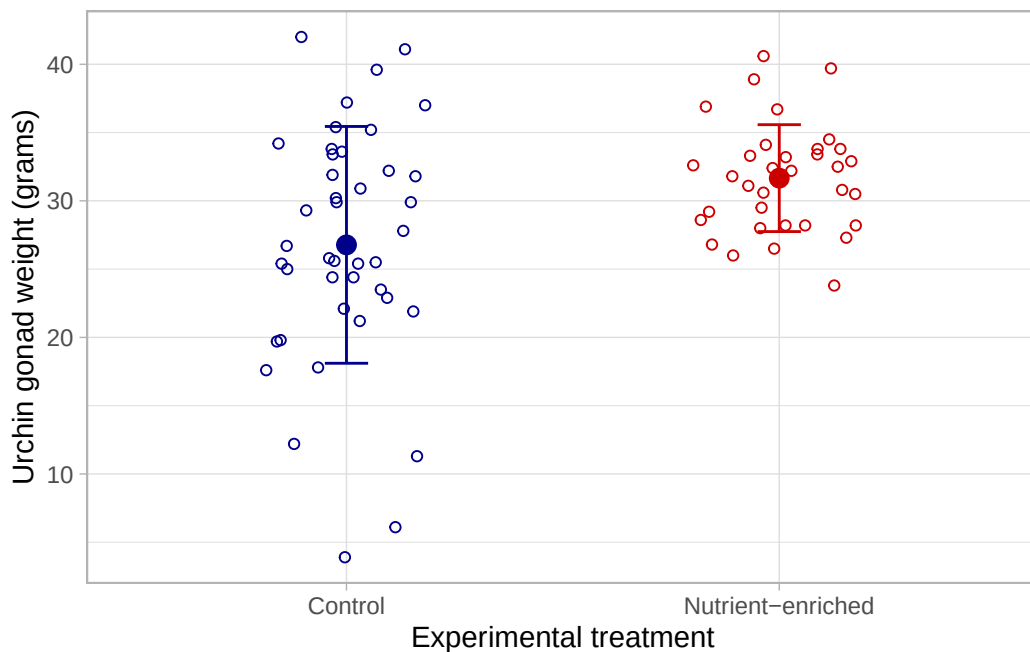
c.

```
# start with the urchins_clean data frame
ggplot(data = urchins_clean,
  # x-axis should be experimental treatment
  # y-axis should be urchin gonad weight
  # coloring by experimental treatment
  mapping = aes(x = experimental_treatment,
    y = weight_g,
    color = experimental_treatment)) +
  # first layer: jitterplot
  geom_jitter(height = 0, # no jitter in vertical direction
    width = 0.2, # small horizontal jitter
```

```

    shape = 21) + # open circle shape
# second layer: represent means at each site
stat_summary(geom = "point",
             fun = mean,
             size = 3) + # make points bigger
# third layer: bars to represent standard deviation
geom_errorbar(data = urchins_summary, # use summary data frame
             mapping = aes(x = experimental_treatment,
                           y = mean, # assign axes based on summary frame
                           ymin = mean - sd,
                           ymax = mean + sd,
                           width = 0.1)) + # make the bars narrower
# relabelling x- and y-axes
labs(x = "Experimental treatment",
     y = "Urchin gonad weight (grams)") +
theme_light() + # custom theme
# custom colors
scale_color_manual(values = c("Control" = "darkblue",
                              "Nutrient-enriched" = "red3")) +
# taking out legend
theme(legend.position = "none")

```



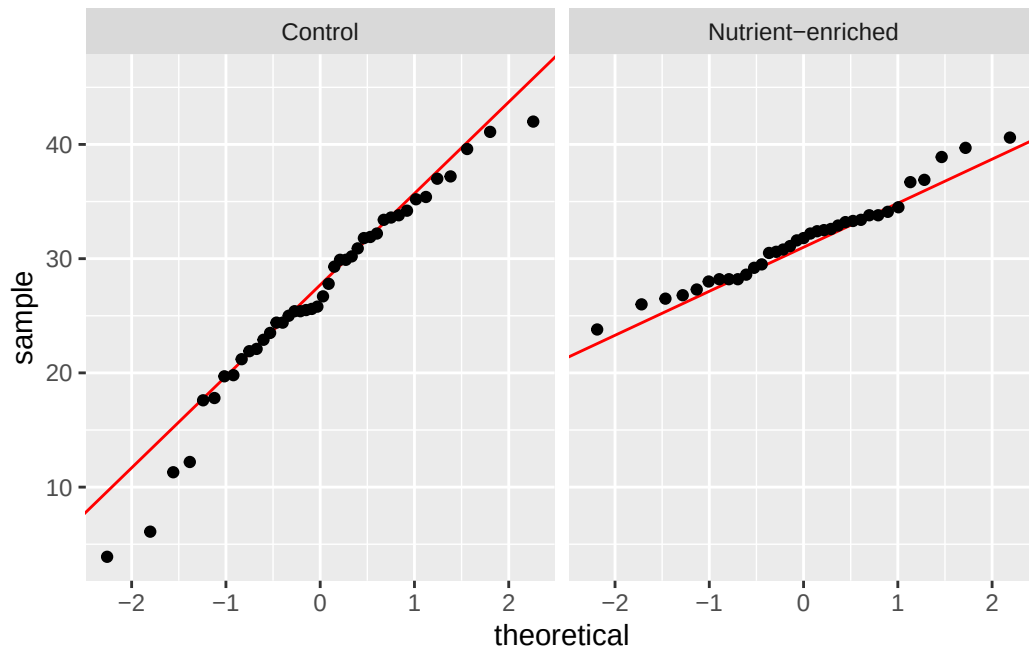
d. Null hypothesis: The difference in mean urchin gonad weight between urchins who

consume normal seaweed and those who consume nutrient-enriched seaweed is equal to 0.

Alternative hypothesis: The difference in mean urchin gonad weight between urchins who consume normal seaweed and those who consume nutrient-enriched seaweed is not equal to 0.

e.

```
# base layer: ggplot
ggplot(data = urchins_clean, # starting data frame
       aes(sample = weight_g)) + # y-axis for QQ plot
# first layer: QQ reference line
geom_qq_line(color = "red") + # adding a color
# second layer: QQ plot
geom_qq() +
# creating "facets"
facet_wrap(~experimental_treatment) # show treatments separately
```



f. Urchin gonad weight is normally distributed because the data points follow a straight diagonal line with a 45 degree angle.

g.

```
# calculate variances with gut check
urchin_weight_var <- urchins_clean |> # starting data frame
  group_by(experimental_treatment) |> # group by experimental treatment
  summarize(variance = var(weight_g)) # calculate variances

# calculate variance ratio
urchin_weight_var
```

```
# A tibble: 2 x 2
  experimental_treatment variance
  <chr>                <dbl>
1 Control                75.2
2 Nutrient-enriched     15.3
```

```
75.2/15.3 # gut check
```

```
[1] 4.915033
```

```
# statistical check: F test of equal variances
var.test(
  weight_g ~ experimental_treatment, # formula: response variable ~ grouping variable
  data = urchins_clean) # data: urchins-clean data frame
```

F test to compare two variances

```
data: weight_g by experimental_treatment
F = 4.9054, num df = 41, denom df = 34, p-value = 7.185e-06
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 2.527793 9.330193
sample estimates:
ratio of variances
 4.905405
```

h.

```
t.test(
  # formula: response variable ~ grouping variable
  weight_g ~ experimental_treatment,
  # argument for unequal variances
  var.equal = FALSE,
  # data: urchins_clean data frame
  data = urchins_clean)
```

Welch Two Sample t-test

```
data: weight_g by experimental_treatment
t = -3.2739, df = 59.238, p-value = 0.001774
alternative hypothesis: true difference in means between group Control and group Nutrient-
enriched is not equal to 0
95 percent confidence interval:
 -7.873119 -1.900214
sample estimates:
      mean in group Control mean in group Nutrient-enriched
      26.77619                      31.66286
```

- i. A t-test would have been appropriate for testing my hypothesis from part d because I am testing for a true difference in means between two groups. I evaluated normality by creating a QQ plot and evaluated homogeneity of variance by manually calculating the variances and running an F test. Because the group variances were unequal, I decided to do a Welch's Two Sample t-test.

j.

```
# calculate cohen's d
d <- cohen.d(weight_g ~ experimental_treatment, data = urchins_clean)

d # display effect size value
```

Cohen's d

```
d estimate: -0.7049911 (medium)
95 percent confidence interval:
      lower      upper
-1.1747569 -0.2352252
```

- k. We found evidence to suggest that consumption of nutrient-enriched seaweed has a moderate effect (Cohen's $d = 0.70$) on mean urchin gonad weights (Welch Two Sample t-test, $t(59.2) = -3.27$, $p=0.002$, $\alpha = 0.05$). On average, urchins who ate nutrient-enriched seaweed ($n=35$, $\text{mean}=31.8$) had higher gonad weights than urchins who ate normal seaweed ($n=42$, $\text{mean}=26.7$).

Problem 3. Personal data

a.

```
# cleaning data
plans_and_duration <- rides |> # create new object from rides
  clean_names() |> # clean column names
# select plans and duration columns
select(plans_after_class, duration_seconds)

# create new object from plans_and_duration
plans_and_duration_summary <- plans_and_duration |>
  group_by(plans_after_class) |>
  summarize( # what I want in the summary
    mean = mean(duration_seconds),
    n = length(duration_seconds),
    sd = sd(duration_seconds))

plans_and_duration_summary # display summary
```



```
# A tibble: 2 x 4
  plans_after_class mean      n    sd
  <chr>             <dbl> <int> <dbl>
1 no                379.     3  11.6
2 yes                378.     2  21.9
```



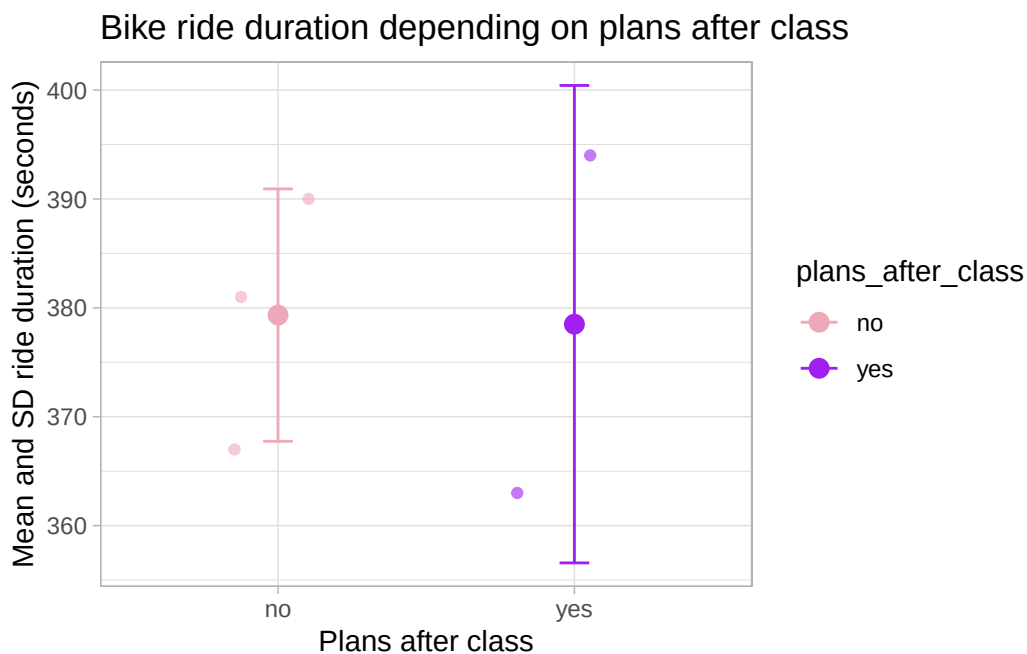
```
# base layer ggplot
ggplot(data = plans_and_duration,
  # x-axis should be plans after class
  mapping = aes(x = plans_after_class,
    y = duration_seconds, # y-axis is ride duration
    color = plans_after_class)) + # color by plans status
# first layer: showing the underlying data
geom_jitter(height = 0, # no jitter in the vertical direction
  width = 0.2, # smaller jitter in the horizontal direction
```



```

    alpha = 0.6) + # make the points more transparent
# second layer: represent means at each site
stat_summary(geom = "point",
             fun = mean,
             size = 3) + # make point bigger
# third layer: bars to represent standard deviation
geom_errorbar(data = plans_and_duration_summary, # use summary data frame
             mapping = aes(x = plans_after_class,
                           y = mean, # assign axes based on data frame
                           ymin = mean - sd,
                           ymax = mean + sd),
             width = 0.1) + # make the bars narrower
# relabelling x- and y-axes, adding descriptive title
labs(title = "Bike ride duration depending on plans after class",
     x = "Plans after class",
     y = "Mean and SD ride duration (seconds)") +
theme_light() + # custom theme
scale_color_manual(values = c("yes" = "purple", # custom colors
                              "no" = "pink2"))

```



b.

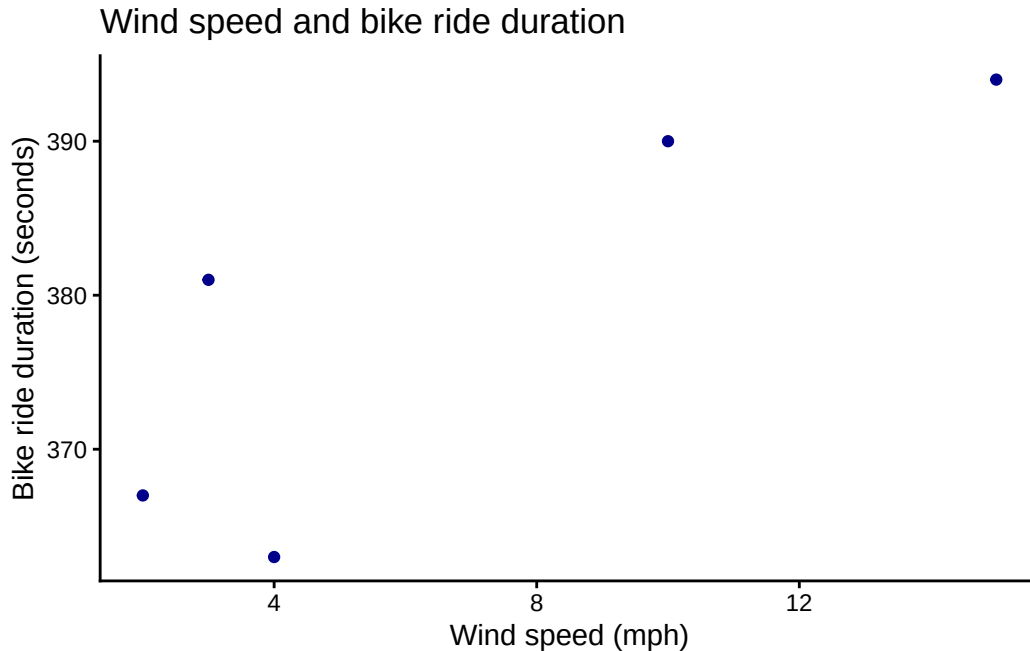
```

# cleaning data
wind_speed_and_duration <- rides |> # create new object from rides
  clean_names() |> # clean column names
# select wind speed and duration columns
select(wind_speed_mph, duration_seconds)

# plot the data, base layer ggplot
ggplot(data = wind_speed_and_duration,
  # assign x as wind speed, y as ride duration
  mapping = aes(x = wind_speed_mph,
    y = duration_seconds,
    color = "wind_speed_and_duration")) + # add color

# first layer: points
geom_point() +
# manually assign colors
scale_color_manual(values = c("wind_speed_and_duration" = "darkblue")) +
  # rename x and y axes, add descriptive title
labs(title = "Wind speed and bike ride duration",
  x = "Wind speed (mph)",
  y = "Bike ride duration (seconds)") +
theme_classic() + # custom theme
# taking out legend
theme(legend.position = "none")

```



- c. After analyzing my visualization in part a, I found that having plans after class might not correlate with a shorter bike ride duration, as the points representing the group means have very similar placement in comparison to the y-axis. As I collect more data and update the figure from a, the means might diverge if having plans would cause me to bike faster. From my visualization in part b, I found that higher wind speeds may correlate with longer bike ride durations, as there is an overall positive direction to the line graph. As I collect more data and update the figure from b, the line graph may look less like a zig-zag and conform more closely to a single slope.
- d. When creating my first visualization, I found that the duration data that I had in mm:ss form did not translate into a readable format on the graph. This challenge arose because I was trying to graph it as a simple numeric value, but the data was not in a simple numeric form. To fix this, I went to my original data spreadsheet and converted the duration column from mm:ss to seconds, as well as relabeling the column's units. I then updated the file in my folder and updated the column name in my code.

Problem 4. Statistical Critique

- a. Why were you interested in this paper?

I have never learned about the effects of young animals spending time with their mothers before, and the goal of the paper was easy for me to visualize.

- b. What questions/hypotheses are the authors addressing?

The authors address whether increased contact between calves and the herd, horned and without horns, has an association with the calves' growth and health.

- c. What is the name of the test you have selected? What is the response variable? What is the predictor variable?

I selected the Kruskal-Wallis test. The response variable is number of veterinary treatments received, and the predictor variable is time the calf spent with the herd.

- d. How does this test address the main question(s) presented by the authors? For example, how would the authors interpret a "significant" result ($p < 0.05$)?

This test was used only to determine whether there was a difference in number of veterinary treatments received by calves depending on the time they spent with the herd, corresponding with the main question of calf health. The authors would interpret a significant result as a difference in the number of veterinary treatments given to calves who spent different amounts of time with the herd.

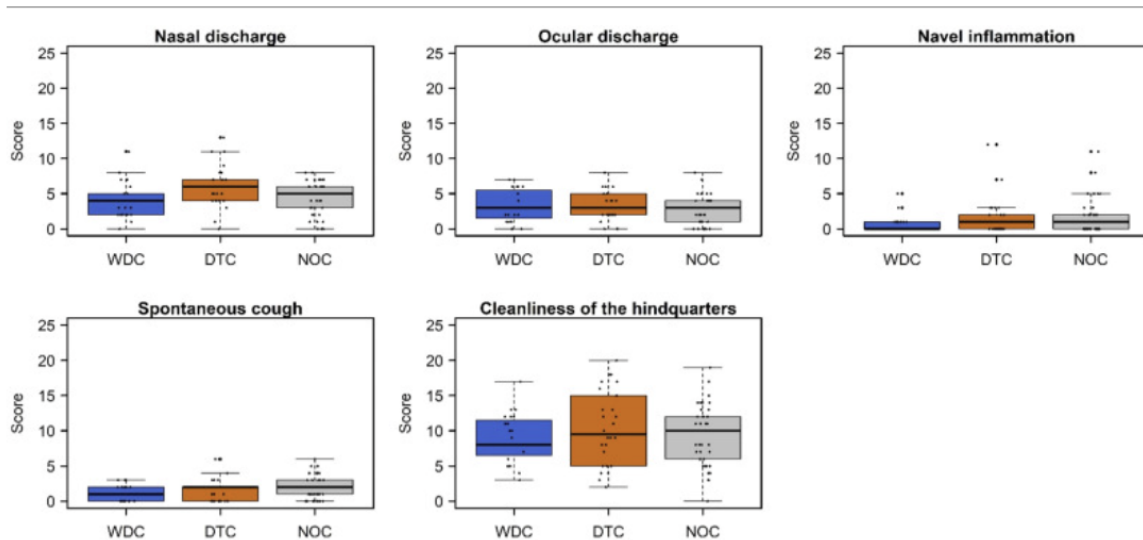


Figure 1: Kruskal-Wallis Test Figure

- e.
- f. The figure includes five different box plots, each one specific to a clinical symptom.

x-axis: calf group (WDC - whole-day contact, DTC - daytime contact, NHC - no herd contact)

y-axis: Health parameter score (0-25)

The figure shows the relationship between the amount of time that calves spent with the herd and their health parameter scores spanning five clinical symptoms.